BIRD Router Verification Test Plan

Directed Tests

Test Name	Purpose	Expected Result	Coverage
local_basic_test	Check a basic valid local packet	Local output matches payload and CRC	Local traffic, valid local cfg, payload_len == 1
local_multi_byte_test	Check local packet with multiple payload bytes	Local output forwards all payload bytes correctly	Local traffic, payload_len 2..4
local_seq_non1_valid_test	Check local packet with valid non-1 sequence number	Packet is accepted and forwarded correctly	Local traffic, valid local cfg, seq_num
local_seq_max_valid_test	Check local packet with maximum valid sequence number	Packet is accepted and forwarded correctly	Local traffic, valid local cfg, seq_num boundary
local_payload_len_large_valid_test	Check local packet with large valid payload length	Full local payload is forwarded correctly	Local traffic, payload_len > 4
local_seq_zero_drop_test	Check local packet with SEQ_NUM=0	Packet is dropped and drop_cnt increments	Local seq zero, drop count
local_frag_zero_drop_test	Check local packet with invalid FRAG_NUM=0	Packet is dropped and no local output is produced	Invalid local frag, drop count
local_invalid_frag_test	Check local packet with invalid local fragment number	Packet is dropped	Invalid local frag, drop count
payload_len_zero_drop_test	Check local packet with payload_len=0	Packet is dropped and drop_cnt increments	payload_len == 0, drop count
remote_payload_len_zero_drop_test	Check remote packet with payload_len=0	Packet is dropped and drop_cnt increments	payload_len == 0, remote traffic, drop count
remote_single_fragment_test	Check valid remote packet with a single fragment	Remote output matches expected data and CRC	Remote traffic, valid remote cfg, remote single fragment
remote_multi_byte_test	Check remote packet with multi-byte payload	Remote output words match expected payload and CRC	Remote traffic, payload_len 2..4
remote_payload_len_large_valid_test	Check remote packet with large valid payload	Remote output matches expected large payload behavior	Remote traffic, payload_len > 4
remote_seq_max_valid_test	Check remote packet with maximum valid sequence number	Packet is accepted and handled correctly	Remote traffic, valid remote cfg, seq_num boundary

remote_seq_zero_drop_test	Check remote packet with SEQ_NUM=0	Packet is dropped and drop_cnt increments	Remote seq zero, drop count
remote_frag_zero_drop_test	Check remote packet with FRAG_NUM=0	Packet is dropped and drop_cnt increments	Remote frag zero, drop count
remote_two_fragment_in_order_test	Check two remote fragments arriving in order	Fragments are merged into correct remote output	Remote two fragments in order, valid remote cfg
remote_two_fragment_out_of_order_test	Check two remote fragments arriving out of order	DUT behavior is checked against expected fragment ordering	Remote two fragments out of order
remote_three_fragment_in_order_test	Check three remote fragments arriving in order	Fragments are merged correctly	Remote three fragments
remote_three_fragment_3_1_2_test	Check three remote fragments arriving in 3-1-2 order	DUT behavior is checked for out-of-order fragment merge	Remote three fragments, fragment ordering
remote_seq_mismatch_drop_test	Check remote sequence mismatch while packet is incomplete	Packet is dropped or mismatch is reported	Remote seq mismatch drop, drop count
remote_seq_mismatch_same_frag_drop_test	Check remote sequence mismatch using same fragment position	Packet is dropped or mismatch is reported	Remote seq mismatch drop
remote_new_packet_while_incomplete_drop_test	Check new remote packet arriving while previous packet is incomplete	Previous incomplete packet is dropped and drop_cnt increments	Remote new packet while incomplete drop
reserved_bits_drop_test	Check packet with reserved configuration bits set	Packet is dropped and drop_cnt increments	Reserved bits error, drop count
reserved_high_bits_drop_test	Check packet with high reserved bits set	Packet is dropped	Reserved bits error, drop count
remote_reserved_high_bits_drop_test	Check remote packet with high reserved bits set	Packet is dropped	Reserved bits error, remote traffic, drop count
remote_reserved_mid_bits_drop_test	Check remote packet with middle reserved bits set	Packet is dropped	Reserved bits error, remote traffic, drop count
drop_count_accumulation_test	Check multiple invalid packets in sequence	drop_cnt accumulates correctly	Drop count observed
drop_count_hold_after_valid_test	Check that drop count holds after a valid packet	drop_cnt does not reset incorrectly after valid traffic	Drop count observed, valid traffic after drop

local_backpressure_test	Check local output when receiver applies backpressure	Local output data remains correct and stable	Local backpressure
remote_backpressure_test	Check remote output when receiver applies backpressure	Remote output data remains correct and stable	Remote backpressure
reset_initial_test	Check reset behavior at the start of simulation	Outputs and counters reset to initial values	Reset initial
reset_after_drop_test	Check reset behavior after a dropped packet	Drop state is cleared after reset	Reset after drop
reset_during_remote_accumulation_test	Check reset while a remote packet is incomplete	Incomplete remote packet is cleared and normal traffic works after reset	Reset during remote accumulation
local_remote_switching_test	Check switching between local and remote traffic	Both local and remote paths operate correctly	Local traffic, remote traffic, switching
remote_frag2_seq2_next	Check remote next-case fragment/sequence behavior	DUT behavior is checked for this sequence/fragment case	Remote fragment/sequence case
remote_frag3_seq4_next	Check remote next-case fragment/sequence behavior	DUT behavior is checked for this sequence/fragment case	Remote fragment/sequence case
remote_frag5_seq3_next	Check remote next-case fragment/sequence behavior	DUT behavior is checked for this sequence/fragment case	Remote fragment/sequence case

Random Tests

Test Name	Purpose	Expected Result	Coverage
random_all_cases_test	Randomly exercise valid and invalid packet combinations	DUT behavior matches scoreboard expectations for generated scenarios	Local traffic, remote traffic, payload length, drop cases
Random local cases	Randomly check local packet fields and payloads	Valid local packets pass and invalid local packets drop	Local cfg, payload_len, seq_num, frag_num
Random remote cases	Randomly check remote packet fields, fragments, and payloads	Valid remote packets are handled and invalid remote packets drop	Remote cfg, remote seq zero, remote frag zero, payload_len
Random drop cases	Randomly generate invalid packets	drop_cnt increments when invalid packets are dropped	Drop count observed, reserved bits, invalid cfg

Functional Coverage Summary

Area	Functional Coverage Bins
Traffic	Local transactions, remote transactions
Fields	seq_num, frag_num, payload_len, payload
Payload Length	payload_len == 0, payload_len == 1, payload_len 2..4, payload_len > 4
Local Config	Valid local cfg, local seq zero, invalid local frag
Remote Config	Valid remote cfg, remote seq zero, remote frag zero
Errors	Reserved bits error, drop count observed
Backpressure	Local backpressure, remote backpressure
Reset	Reset initial, reset after drop, reset during remote accumulation
Remote Merge	Remote single fragment, two fragments in order, two fragments out of order, three fragments
Remote Drop	Remote seq mismatch drop, remote new packet while incomplete drop